1. What is SQL? Why is it used?

Follow-up Questions:

- Is SQL a programming language?
- Which databases support SQL?
- What are the different categories of SQL commands?

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2. What is the difference between WHERE and HAVING?

Follow-up Questions:

- Which clause is executed first?
- Can HAVING be used without GROUP BY?
- Which clause is used with aggregate functions?

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3. What is the difference between GROUP BY and ORDER BY?

Follow-up Questions:

- Can both be used in the same query?
- Which clause groups records?
- Which clause sorts records?

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4. What is the difference between COUNT(*), COUNT(column), and COUNT(DISTINCT column)?

Follow-up Questions:

- Which one counts NULL values?
- Give an example of each.
- When would you use COUNT(DISTINCT)?

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5. What are Aggregate Functions?

Follow-up Questions:

- Name five aggregate functions.

- Can aggregate functions ignore NULL values?
- Which clause is commonly used with aggregate functions?

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6. What is the difference between UNION and UNION ALL?

Follow-up Questions:

- Which one removes duplicate records?
- Which one is faster?
- What conditions must both SELECT statements satisfy?

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7. What is the difference between a Subquery and a Join?

Follow-up Questions:

- When would you prefer a Join?
- Can a subquery return multiple rows?
- Can a subquery be nested?

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8. What is the difference between CHAR and VARCHAR?

Follow-up Questions:

- Which one saves storage?
- When would you choose CHAR over VARCHAR?
- Give practical examples.

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9. Write SQL queries for the following:

- Display all students.
- Display students whose marks are greater than 80.
- Display students sorted by marks in descending order.
- Count the total number of students.
- Find the maximum salary from the Employee table.

Follow-up Questions:

- Explain each query.
- Can you optimize any of them?

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10. Scenario-Based Question

Consider the following tables:

Employee(EmpID, Name, DeptID, Salary)

Department(DeptID, DeptName)

Write SQL queries to:

- Display employee name along with department name.
- Find the highest-paid employee.
- Display employees whose salary is above the average salary.
- Count employees department-wise.
- Display departments having more than five employees.

Follow-up Questions:

- Which JOIN will you use?
- Can you solve this using a subquery?
- Which solution is more efficient?